MIL-C-915/77B
18 June 1985
SUPERSEDING
MIL-C-915/77A(SH)
25 January 1982

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE 2SWL-7 (INCLUDING VARIATION TYPE 2SWLA-7)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-915.

This cable can be manufactured in two variations: 2SWL-7 (unarmored) and 2SWLA-7 (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, tin coated, AWG no. 16, class B stranding.
Second - Polyethylene insulation, type II, class L, grade 4 in accordance with L-P-390. Nominal thickness 0.016 inch. Identification method 3.
Third - Clear polyamide jacket over the insulation, in accordance with ASTM D 4066. Jacket nominal thickness 0.003 inch.
Fourth - Two insulated conductors (one black, one white), cabled together with a nominal lay of 3 inches, to form a pair. At manufacturer's option, a binder tape applied helically with overlap.
Fifth - Braided shield of tin coated copper, size either AWG no. 34 or AWG no. 36. Braid angle 30 to 35 degrees, minimum coverage of 85 percent.
Sixth - Shield insulation. (Either (a) or (b)):

(a) One polyester tape, type G in accordance with MIL-I-631, plus a jacket of clear polyamide in accordance with ASTM D 4066, nominal thickness 0.003 inch. Standard identification code applied by method 2.
(b) Two polyester tapes, type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2, on inner tape.

FSC 6145

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Seventh - Seven shielded pairs cabled together with a lay not greater than 18 times the overall diameter of the layer. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Filler shall be nonfibrous.
Eighth - Binder tape applied helically with overlap.
Ninth - Special thermoplastic jacket (gray) with cable surface marking. Minimum jacket thickness shall be 0.060 inch. Cable overall diameter shall be 0.860 minimum, 0.910 maximum.

Construction variant for type 2SWL-7

Ninth - Special thermoplastic jacket (gray) with cable surface marking. Minimum jacket thickness shall be 0.060 inch. Cable overall diameter shall be 0.860 minimum, 0.910 maximum.

Construction variant for type 2SWLA-7

Ninth - Special thermoplastic jacket (gray) with cable surface marking. Minimum jacket thickness shall be 0.060 inch. Cable overall diameter shall be 0.860 minimum, 0.910 maximum.
Tenth - Braided metal armor and paint.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	4.44
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield (pair shields only)	500
Shield to water (2SWL-7 only)	500
Insulation resistance - megohms/1000 feet, minimum	
Conductor to conductor	3000
Conductor to shield	1500
Shield to shield	100
Shield to water (2SWL-7 only)	500
Shield to armor (2SWLA-7 only)	500
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-915, table XXII, for limits of water leakage	No failure
Mutual capacitance - at 1 kHz, pF/ft, maximum	35.0
Capacitance unbalance - percent, maximum	10

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group B:	
Drip - at 95 ± 1°C	Zero
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - lbf/in^2, minimum	1400
Elongation - percent, minimum	300
Jacket (cable)	
Tensile strength - lbf/in^2, minimum	2300
Elongation - percent, minimum	250
Group C:	
Physicals (aged)	
Insulation (conductor)	
Air oven	
Elongation - percent of unaged, minimum	60
Jacket (cable)	
Air oven (168 hours at 136°C)	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged, minimum	50
Permanence of printing (jacket) type 2SWL - cycles, minimum	250
Cable filler removability	No failure
Armor - conformance to materials, construction and coverage, type 2SWLA	No failure
Shield-conformance to material coverage and construction	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Cable aging and compatibility (95 ± 3°C)	No failure
Jacket durometer (type A) hardness, minimum	80

UNIT ORDERING LENGTH: 1500 feet (nominal).

Revision letters are not used to denote changes due to the extensiveness of the changes.

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NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

Preparing activity:
Navy - SH
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